Prof. WANG, Juncheng

王俊程教授

BEng, MSc, PhD
Assistant Professor, Department of Computer Science

jcwang@comp.hkbu.edu.hk

DLB 633 +852 3411 5998

Personal Webpage HKBU Scholars

About

Juncheng Wang received the Ph.D. degree in Electrical and Computer Engineering from the University of Toronto, Toronto, ON, Canada, in 2023, the M.Sc. degree in Electrical and Computer Engineering from the University of Alberta, Edmonton, AB, Canada, in 2017, and the B.Eng. degree in Electrical Engineering from Shanghai Jiao Tong University, Shanghai, China, in 2014. His current research interests are computer and communication networks.

Research Interests

- Network Artificial Intelligence
- Mobile Communications
- Online Learning
- Distributed Computing
- Stochastic Optimization

Perspective Students: I am always looking for self-motivated Ph.D. students (multiple openings), Research Assistants, Post-docs, and Visitors to work with me (see this document for more details). If you find there are some common research interests between us, please feel free to drop me an email with your CV, transcripts, and any other supporting materials.

Selected Publications

My Students

- W. Qin, W. Bao, **J. Wang***, and M. Zhou, "Double queue for constrained online convex optimization: Bridging the best-of-two-worlds constraint violation," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2026.
- **J. Wang***, Y. Liu, B. Liang, and M. Dong, "Constrained over-the-air model updating for wireless online federated learning with delayed information," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2025.
- **J. Wang***, B. Yan, and Y. Liu, "Doubly-bounded queue for constrained online learning: Keeping pace with dynamics of both loss and constraint," in *Proc. AAAI Conf. Artif. Intell. (**AAAI**)*, 2025. (Theory paper)
- **J. Wang***, B. Liang, M. Dong, G. Boudreau, and A. Afana, "Age-of-information minimization with weight limits for semi-asynchronous online distributed optimization," *IEEE Trans. Netw. (**TON**)*, 2025.
- **J. Wang**, M. Dong, B. Liang, G. Boudreau, and A. Afana, "Exploring temporal similarity for joint computation and communication in online distributed optimization," *IEEE Trans. Netw. (**TON**)*, 2025.
- **J. Wang**, B. Liang, M. Dong, G. Boudreau, and H. Abou-Zeid, "Joint online optimization of model training and analog aggregation for wireless edge learning," *IEEE/ACM Trans. Netw. (**TON**)*, 2024.

The above papers are published after I joined HKBU.

- **J. Wang**, M. Dong, B. Liang, and G. Boudreau, "Periodic updates for constrained OCO with application to large-scale multi-antenna systems," *IEEE Trans. Mobile Comput. (**TMC**)*, 2023.

- **J. Wang**, B. Liang, M. Dong, G. Boudreau, and A. Afana, "Online distributed optimization with efficient communication via temporal similarity," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2023.
- **J. Wang**, M. Dong, B. Liang, G. Boudreau, and H. Abou-Zeid, "Delay-tolerant OCO with long-term constraints: Algorithm and its application to network resource allocation," *IEEE/ACM Trans. Netw. (**TON**)*, 2023.
- **J. Wang**, M. Dong, B. Liang, G. Boudreau, and H. Abou-Zeid, "Online model updating with analog aggregation in wireless edge learning," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2022.
- **J. Wang**, B. Liang, M. Dong, G. Boudreau, and H. Abou-Zeid, "Semi-online precoding with information parsing for cooperative MIMO wireless networks," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2022.
- **J. Wang**, B. Liang, M. Dong, G. Boudreau, and H. Abou-Zeid, "Delay-tolerant constrained OCO with application to network resource allocation," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2021.
- **J. Wang**, M. Dong, B. Liang, and G. Boudreau, "Online precoding design for downlink MIMO wireless network virtualization with imperfect CSI," in *Proc. IEEE Conf. Comput. Commun. (**INFOCOM**)*, 2020.